

2. Sensors for voltage, current, temperature, and vibration
3. Controlled switches (MOSFET/IGBT/relays/contactors)
4. Indicators (optional)
5. Communication peripherals, either as part of the MCU or a separate IC

Let us first understand the various aspects that need to be controlled/monitored, so that we can later on assign each of these jobs to specific entities of our hypothetical BMS. Before we start, remind yourself of the adage, *Design is all about choices and trade-offs!*

A typical connection diagram of a 4S3P battery type is shown with its BMS in Fig. 1. Please notice the following important things here:

1. The BMS is an integral part of the battery. Unprotected Li-ion batteries are unsafe and unusable
2. The BMS is connected to each cell pack (cells in parallel) in series
3. The BMS controls the charging and discharging of the battery pack
4. As the number of series string cell packs increases, so does the BMS complexity

Monitoring health of the battery

Battery health has multiple parameters, the most essential of which are its terminal voltage, charge capacity, and cell temperature. So, monitoring battery health involves monitoring these parameters continuously.

Voltage monitoring. A battery, as the name suggests, comprises individual electrochemical cells that are connected in series or parallel. So, each cell voltage should also be monitored. Let us understand what the cell voltage represents.

The specific energy (or energy density or energy per unit volume) of a particular cell chemistry is unique to its molecular composition. This specific energy directly affects the terminal voltage of a cell. Thus, higher the terminal voltage of a particular chemistry, higher is its specific energy.

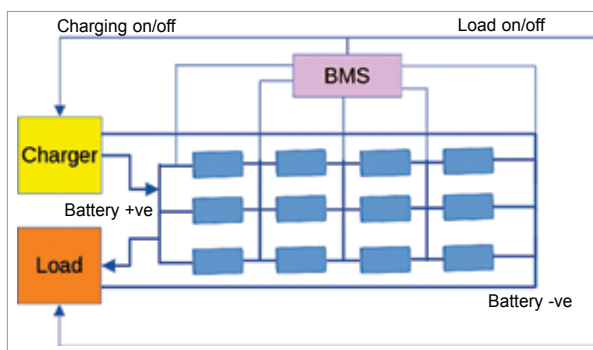


Fig. 1: Typical connection of a BMS to a 4S3P style battery

To sustain the reversible nature of the cells' electrochemical properties, it is necessary that the energy content in the cell should not fall below a certain level or rise above a certain level. In simpler words, to keep the cells healthy, and in turn the battery healthy, it is essential to keep the cells' terminal voltage within a given voltage envelope.

This safe voltage envelope (SVE) is different for every chemistry. For example, the lead-acid chemistry has an open circuit voltage (OCV) of approximately 2.1V and an SVE of 1.8V to 2.35V. The lithium-iron-phosphate (LiFePO₄) cell has an OCV of approximately 3.2V and an SVE of 2.5V to 3.65V.

Since each cell of a battery is made individually, and because it is impossible to make completely identical cells in a batch or across batches, nominal cell voltage of each cell may be different inside the battery. Thus, if a battery has 100 cells, would the designer have to monitor voltage for all the cells? Just ponder over it. We shall try to answer this while looking at energy redistribution strategies later in this article.

Charge capacity monitoring.

The charge capacity of the cell is of primary importance when cell health is considered. The interesting part of the charge capacity is that for most chemistries it is reflected in some part as the terminal voltage of the cell.

In case of Li-ion cells, the change in terminal voltage with respect to its charge capacity is quite flat.

This means that the terminal voltage does not change much with reduction in charge capacity. Thus, the depth of discharge (DoD) levels or the state of charge (SoC) levels cannot be faithfully calculated all along their charge/discharge curve based solely on the terminal voltage.

If we consider the discharge curve of a Li-ion cell, major changes in the OCV of the cell occur only at the extreme ends, and all through the curve the OCV almost stays flat with change in the DoD. This is in stark difference from lead-acid cells, where the OCV changes quite a bit with increasing DoD.

This unique nature of the Li-ion cells means that a BMS must use an advanced approach to ascertain the running DoD levels. The typical BMS achieves this by using multiple techniques, such as the use of high granularity analogue front-end and using Coulomb Counting to track the charging and discharging levels and rates. Some advanced BMSes may also employ a square wave load at a frequency of 1Hz to 1kHz, to get an idea of the internal resistance of the battery and thereby infer its state of charge.

Cell temperature monitoring. Another parameter to monitor is the cell temperature. Cell temperature affects the electrochemical reactions and intercalation processes in the cell. An excessively high cell temperature can result in gasification in lead-acid cells and can result in a thermal runaway failure in Li-ion cell chemistries. An excessively low temperature can permanently damage a Li-ion cell.

This means that a BMS should monitor cell temperature all the time. But doing this for each individual cell would be quite sub-optimal for the overall battery design. This is where the BMS designer must take a call on placement of temperature